

GAEA-Net: Generating Activity-Enriched Abnormal ECGs via Adversarial Network

Liuqing Chen, Shuhong Xiao, Yujie Zang, Jiner Wang, Shanhai Hu, Lin Zhang, Qing Li, Danyang Zhang

Abstract—With the increasing demand for personalized health monitoring through wearable devices, there is a growing need for non-prescription ECG diagnosing, especially during physical activities. However, existing abnormal ECG data are typically measured in clinical settings, reflecting heart waveforms in a resting state. Abnormality classification models based on such data often struggle to maintain high performance during physical activities, leading to increased false alarms and a higher incidence of missed detections. Due to the potential risks associated with having patients engage in physical activity, abnormal ECG data captured during exercise is not readily available, further complicating the development of reliable models for active scenarios. To address this issue, we propose GAEA-Net in this study. Our goal is to utilize exercise ECGs from healthy individuals, which are more easily accessible, combined with resting-state abnormal ECGs, to generate activity-enriched ECGs through synthesis. We conduct abnormal classification on five widely used datasets, achieving average improvements of 1.3% in Accuracy, 1.3% in F1-score, 0.9% in AUROC, 1.6% in MCC, and 1.4% in Cohen's Kappa. Furthermore, a clinical Turing test involving seven experienced cardiologists confirms that our synthesized ECGs exhibit high fidelity. In the diagnostic task, the cardiologists achieved comparable accuracy on synthetic and real ECGs (55.7% vs. 54.9%, $p = 0.76$). Code and data will be made available at <https://github.com/XiaoShuhong/GAEA-NET> later.

Index Terms—ECG Synthesis, Generative Adversarial Network, Abnormal ECG Classification, Clinical Turing Test, Activity Level Enrichment

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INTRODUCTION

As a non-invasive diagnostic tool, electrocardiogram (ECG) has been extensively utilized in clinical settings for detecting cardiac abnormalities [1]. Recent advancements in deep learning have significantly enhanced ECG analysis, enabling classification models [2]–[4] to deliver expert-level accuracy in preliminary cardiovascular disease classification while streamlining diagnostic workflows for cardiologists in clinical practice. Meanwhile, the advent of wearable device and heightened demand for personalized health monitoring are expanding automated diagnosis into daily-life applications [5], [6], necessitating models that address challenges distinct from controlled clinical environments. Specifically, physiological variability during physical activity introduces complexities beyond those encountered in standardized recordings.

Empirical studies [7], [8] have demonstrated that exercise-induced hemodynamic changes can significantly alter ECG morphology, such as ST-segment depression. However, existing datasets [9], [10] primarily capture ECGs in clinical settings, where patients are measured under resting conditions (i.e., at low activity levels) to reduce diagnostic complexity. As a consequence, abnormality detection models trained on such resting-state data often fail to perform adequately during exercise. The exercise-induced artifacts act as outliers, leading to increased false alarms and a higher incidence of missed detections. This data gap remains because collecting exercise ECGs from cardiac patients is both costly and risky [11], and so far, no such datasets have been made publicly available.

Inspired by recent advances in ECG generation [4], [12]–[14], we address this limitation by synthesizing abnormal ECGs that reflect exercise-induced changes. While prior work has focused mainly on disease-induced morphological changes, we go beyond modeling disease effects only by also incorporating non-pathological, activity-related influences such as those induced by physical exertion, enabling the generation of ECGs that more faithfully represent abnormal signals during physical activity. In this context, our task can also be viewed as a domain adaptation problem, where we transform abnormal ECGs from the rest domain (low activity level) to the exercise domain (high activity level). In this task, there are two key challenges. The first challenge is how to effectively extract the activity-related features alongside the health condition features that characterize specific abnormalities. Generally, health condition features are considered domain-invariant and tend to exhibit locality, with different segments reflecting specific regions of the heart [14], [15]. However, they are

often influenced by subject-specific variability, which can lead to considerable differences in how the same abnormality is manifested across individuals. By contrast, activity-related features are domain-specific, manifesting a global influence on the signal [16]. Therefore, it is essential to design a specialized approach that extracts both the domain-invariant health condition features and the domain-specific activity-related features. The second challenge lies in how to inject activity-related features effectively. Most existing methods rely on global feature injection [14]; however, in the case of ECG signals, the impact of physical activity is not uniformly distributed across the waveform. For example, previous clinical studies [7] have shown that the ST segment is one of the most affected position during physical activity. Therefore, a position-aware injection mechanism is better suited to capture the effects of activity, enabling more realistic simulation of exercise-induced changes.

To address these challenges, we propose GAEA-Net, which generates activity-enriched abnormal ECGs through adversarial learning framework. Our approach consists of three main components, as shown in Figure 1. To disentangle the features, we introduce a coupled encoder that jointly learns health condition features, which represent disease-related information, and activity level features, which capture individual-specific characteristics. Subsequently, randomly selected exercise-rest ECG pairs from other healthy individuals serve as references to guide the learning of high activity level features that represent physical exercise states. The generator is tasked with injecting the exercise-related features. Implemented as an upsampling process, it enables dynamic injection of activity-related information through our proposed position-aware instance normalization (PosIN) during signal reconstruction. The discriminator is designed to evaluate signals in both the time and frequency domains, distinguishing real from generated signals.

We evaluate abnormal ECG classification performance on five datasets: PTB-XL [10], MuseECG [17], MedaCare-XL [18], CinC 2017 [19], and PTB dataset [20]. Eight other state-of-the-art (SOTA) generative approaches are included for comparison. For each method, the synthesized data are combined with the original training set as data augmentation, and classification performance is evaluated using a standard classifier [21]. Overall, our method achieves the highest performance across all settings, demonstrating the effectiveness of incorporating activity-related features. A clinical Turing test with seven experienced cardiologists further confirmed the high fidelity of the synthesized signals and their consistency with clinically meaningful patterns.

The contributions of this study are summarized as follows:

- While previous studies primarily focused on generating ECG signals by modeling pathological factors that affect waveform morphology, our approach also considers activity level as a non-pathological factor, enabling the synthesis of ECGs that reflect both health conditions and individual physical states.
- We propose GAEA-Net, a generative adversarial framework designed to disentangle disease-specific and activity-specific features and to enable effective injection.

This design allows us to combine resting abnormal ECGs from patients with exercise ECGs from healthy subjects, facilitating the generation of scarce exercise-state abnormal ECGs.

- We validate the effectiveness of our approach against eight other state-of-the-art methods on five clinical ECG datasets, and further demonstrate the high fidelity of the synthesized signals through a clinical Turing test involving seven experienced cardiologists.

RELATED WORK

Generative Methods for Electrocardiograms

Generative methods for synthesizing electrocardiograms (ECGs) have gained significant attention, aiming to provide large-scale, readily accessible training data for automated ECG analysis [22], [23]. This approach reduces the need for manual annotation and addresses potential privacy concerns. Unlike conventional medical approaches that require anatomically-specific ventricular-torso models to simulate ECGs [18], generative adversarial frameworks allow deep learning practitioners to directly learn data distributions from real ECG signals. Zhu et al. [24], as one of the initial work, employed a BiLSTM-CNN generator within a standard adversarial training setup [25], while Hossain et al. [26] further introduced label conditioning and an auxiliary classification task in the discriminator to counter adversarial examples. Similarly, SimGANs [12] incorporated an Euler loss into the discriminator, inspired by early attempt to construct cardiac dynamical models using ordinary differential equations [27]. Several works also focused on specific vital details in enhancing generation quality. For instance, CardioGAN [13] designed an attention mechanism to capture intricate inter-dependencies among the various parts of real ecg segments. Kuznetsov et al. [28] performed their synthesis on 25 interpretable features, whereas Golany and Radinsky proposed PGANs [4], which learn GAN models per patient and introduce personalized wave constraints to accommodate significant inter-patient variability. Recently, diffusion strategies have also been explored for ECG generation. For instance, Adib et al. [29] introduced a approach based on Improved Denoising Diffusion Probabilistic Models and Wasserstein GAN, while Alcaraz and Strodthoff [30] employed State Space Models. Moreover, several notable works broadened the scope of ECG synthesis. Electrocardio Panorama [15] and ME-GAN [1] investigated multi-view generation, and Auto-TTE [31] demonstrated the feasibility of synthesizing signals from electronic health records. Sarkar and Etemad [32] presented cross-modal transformations between ECG and Photoplethysmogram (PPG), whereas HQ-DCGAN [33] proposed a quantum-device implementation. Finally, Hu et al. [14] introduced a digital-twin approach that generates anomalous signals for otherwise healthy samples.

Exercise Electrocardiograms and Its Applications

Clinical electrocardiograms are typically collected in medical centers under resting conditions, recording patients' cardiac electrical potentials [10]. By contrast, an exercise ECG—also known as cardiac stress testing—captures the heart's electrical

activity during physical exertion. Since the 1960s, exercise ECG has been widely employed for diagnosing myocardial ischemia, and its applications have progressively expanded to include the prediction and assessment of myocardial infarction, the diagnosis of coronary artery disease, the prediction of cardiac death events, and the evaluation of arrhythmias and heart rate variability [11], [34].

Common exercise testing methods include treadmill, bicycle ergometer, and the Master two-step test, all of which incrementally increase the exercise load until the subject's heart rate reaches a near-submaximal level. As the workload and myocardial oxygen demand rise, latent hypoxia or ischemia may manifest, yielding ECG abnormalities that are undetectable at rest. These changes are especially pronounced in patients but can also appear in healthy individuals [35]. One of the most conspicuous findings is ST-segment depression [7], and Kligfield *et al.* [8] have offered deeper insights into the complex nature of these variations.

Conducting exercise testing carries inherent risks, particularly for patients with compromised cardiac conditions. Therefore, such tests must be performed under the close supervision of well-trained physicians, with continuous monitoring to promptly prevent and manage adverse events. Despite numerous studies examining exercise ECG in patients with heart disease [11], [36], no corresponding datasets have been made publicly available. Consequently, an alternative is to utilize accessible exercise ECG data derived from healthy populations. For instance, Giovanni *et al.* [37] collected cycle ergometer data from 22 subjects, recording a single-lead ECG across five phases related to the ventilatory threshold, exhaustion, and recovery. Similarly, Zhang *et al.* [38] gathered lead II ECGs from 45 individuals before and after structured exercise, aiming primarily to investigate the stability of ECG as a biometric identifier under exercise-induced interference. Other relevant datasets include Troika [39], Kaggle [40], Sport DB [41], and the datasets published by Agraftoti *et al.* [42], [43]. Most of these collections contain single-lead signals acquired from wearable devices such as wristbands or smartwatches.

METHODOLOGY

Model Overview

We illustrate the overall framework of our method in Figure 1, which comprises three main parts: (a) the feature separator S , (b) the adversarial-based ECG generator G and (c) the discriminator D . Given clinically measured abnormal ECGs X_d annotated with a health condition label y , we encode them into health condition features that reflect the disease-specific information, and activity-level features that capture individual-specific characteristics. A shift in activity level is learned from paired resting ECGs X_r and exercise ECGs X_e collected from other healthy individuals, and subsequently applied to the activity-level features. Our generator then synthesizes abnormal ECGs that reflect a high activity level by conditioning on the health condition features and the shifted activity-level features, while preserving the original health label y . Finally, the discriminator evaluates both the real and synthesized ECG data, with the goal of distinguishing between them. The main symbols used in this section are summarized in Table I.

Symbol	Quantity
X^d, y	abnormal ECGs measured under resting, clinical settings, labeled with health condition label y
X^r, X^e	paired resting and exercise ECGs collected from other healthy individuals
f_{health}, h_{health}^i	health condition features extracted by the encoder; h_{health}^i denotes the intermediate representation at the i^{th} layer
$\hat{f}_{health} = Q(f_{health})$	quantized health condition features obtained via vector quantization
$f_{activity}, h_{activity}^j$	activity level features extracted by the encoder; $h_{activity}^j$ denotes the intermediate representation at the j^{th} layer
z	the activity latent representations utilized in offset editor, $\mathbb{R}^{C_z \times L_z}$
n_l, n_g	local offset vector learned from paired X^r and X^e ; global offset vector obtained via exponential moving average (EMA)
$f_{activity}^*$	high activity level version of $f_{activity}$, obtained via the offset editor
s	activity latent representations utilized in generator, $\mathbb{R}^{L_s}$

TABLE I: Summary of symbols.

Feature Separator

The identification of abnormalities in ECG waveforms is complicated by individual variability—such as gender and age [16]—and by non-pathological factors, particularly physical activity, which is a key focus of this study and can significantly alter ECG morphology. The goal of our feature separator is to disentangle health condition features $f_{health} \in \mathbb{R}^{C \times L}$, which reflect pathological abnormalities, from activity level features $f_{activity} \in \mathbb{R}^{C \times L}$, which capture individual-specific, non-pathological variations present in the ECG waveform. The health condition encoder is implemented as a 34-layer residual network (ResNet-34), which is effective in capturing pathological patterns in ECGs [44]. In contrast, activity-level features are less complex to learn and exhibit more consistent temporal patterns compared to pathological anomalies, we adopt a shallower 18-layer network (ResNet-18) to encode them.

Inspired by [45], we introduce an interconnection between the two encoders to facilitate information exchange, allowing each encoder to refine its feature representations based on complementary insights from the other. Namely, an interconnection between i^{th} layer output of the health encoder and j^{th} layer output of the activity encoder is defined as:

$$InterConnection(i, j) = f(h_{health}^i) \circ h_{activity}^j, \quad (1)$$

where f is a sub-network to align the shapes and $\circ$ denotes a concatenation operation. A classification loss $\mathcal{L}_{clf}$ is applied to both health condition and activity-level features to constrain our encoders. Specifically, an additional linear layer is used as classifier, with health condition classification performed on abnormal label y^d , and activity level classification conducted on rest ECGs and exercise ECGs in a binary setting $\{0, 1\}$.

We then apply quantization to the health condition features to reduce the impact of minor perturbations [46], ensuring more consistent representations and improving the quality of subsequent generation process [47]. Since the latent space of Variational Autoencoders (VAEs) has been shown to possess effective feature disentangling properties [14], [16], with different directions reflecting distinct attributes, we leverage this characteristic to further isolate health information. Specifically, we recognize the latent space of health conditions as a set of N learnable vectors $v_i \in \mathbb{R}^{L_{vq}}$ for $i \in \{1, 2, \dots, N\}$, with each vector capturing a distinct latent factor of disease. To

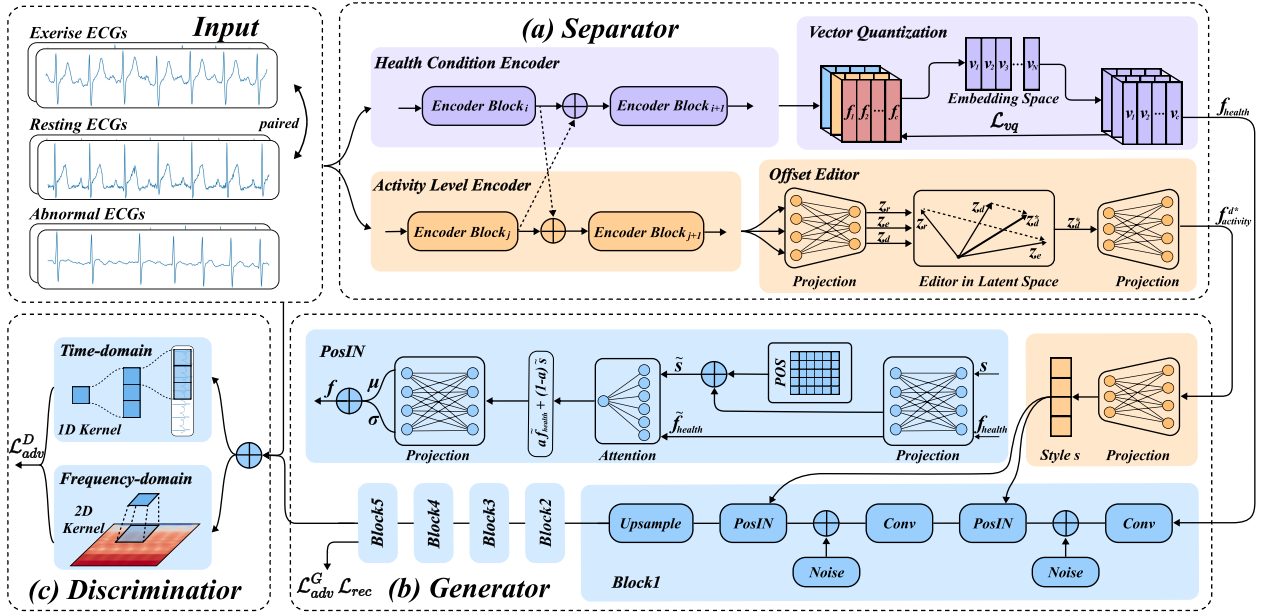


Fig. 1: The framework of our GAEA-Net. We utilize paired exercise and resting ECGs from the same subject combined with clinically measured static abnormal ECGs as synthetic inputs. (a) Illustrating the Separator, in which a health condition encoder with vector quantization is used to obtain the health condition feature, and an activity level encoder with an offset editor is employed to extract the activity level feature. (b) Presenting the generator structure, consisting of a series of upsampling blocks and position-aware instance normalization (PosIN) for activity style injection. (c) Showing the discriminator, distinguishing real from synthetic ECG signals through combined time-domain and frequency-domain branches.

achieve isolation, the quantization process $Q(\cdot)$ is defined as replacing each channel of the health condition features $f_{health} = [f_1, f_2, \dots, f_c]$ as the nearest basis vector v_i :

$$Q(f_c) : \hat{f}_c = v_{\arg \min_i \|f_c - v_i\|^2}. \quad (2)$$

Intuitively, our goal is to retain the common aspects of health features across subjects while discarding minor perturbations. To achieve this, we track the replacement frequency for each learnable vector and update them using an exponential moving average (EMA) instead of backpropagation. Then, the difference between the original f_{health} and quantized $\hat{f}_{health} = Q(f_{health})$ is then used to push the health encoder towards learning more generalizable disease representations. Due to the non-differentiable nature of the quantization process, a stop-gradient operator $sg(\cdot)$ is applied to the quantization loss, defined as:

$$\mathcal{L}_{vq} = \|Q(f_{health}) - sg(f_{health})\|^2, \quad (3)$$

where $sg(\cdot)$ ensures that the value remains unchanged during the forward pass while its gradient is zero during backpropagation.

For the activity level features $f_{activity}$ obtained from our encoder, it currently represents a low activity level under resting conditions. To synthesize abnormal ECGs that exhibit characteristics of high activity levels, we learn a vector offset in the activity space to estimate the expected shift induced by increased physical exertion. Previous research [48] has demonstrated that such a vector space exhibits strong disentanglement properties, allowing for the editing of a single

attribute—activity level in our case—via continuous interpolation, as expressed by:

$$z^* = z + \alpha n, \quad (4)$$

where n is the offset vector and α controls the magnitude of the shift, with z^* , z , and n all lying in the same activity latent space $\mathbb{R}^{C_z \times L_z}$. In the absence of patient-specific ECG data under both rest and exercise, we approximate the offset vector n using matched ECG pairs from healthy subjects. Given X^r and X^e , we extract $f_{activity}^r$ and $f_{activity}^e$ using our activity level encoder, followed by a projection into $\mathbb{R}^{C_z \times L_z}$. The local offset vector n_l is then estimated by concatenating the two projected features and passing them through a series of fully connected layers [48]. As n_l is computed for each mini-batch and may vary significantly, we adopt an EMA to maintain a more stable global offset vector as $n_g \leftarrow \beta n_g + (1 - \beta)n_l$. Subsequently, we perform the continuous interpolation described in Equation (4) using a multi-head cross-attention mechanism, as follows:

$$\begin{aligned} z_d &= proj(f_{activity}), \\ z_d^* &= z_d + MH\text{-Attn}(z_d, n_g), \end{aligned} \quad (5)$$

where z_d is obtained by projecting the activity-level feature $f_{activity}$, and the attention is performed with z_d as the key, and the global offset vector n_g as both the query and value. This allows us to obtain a dynamic, z_d -adaptive weighting of n_g as the expected offset vector. Finally, the shifted high activity level feature is obtained as $f_{activity}^{d*} = W^* z_d + b^*$ through a decoder consisting of a series of fully connected layers. The

loss of our offset editor is designed as a combination of the $\mathcal{L}_{edit}$ and $\mathcal{L}_{reg}$:

$$\begin{aligned}\mathcal{L}_{edit} &= \|f_{activity}^e - f_{activity}^{r*}\|^2, \\ \mathcal{L}_{reg} &= D_{KL}(p(z_d^*) \| p(z_d)),\end{aligned}\quad (6)$$

where $\mathcal{L}_{edit}$ is computed by applying the learned offset back to z^r to generate the edited activity feature $f_{activity}^{r*}$, and then calculating the reconstruction loss between expected $f_{activity}^{r*}$ and real $f_{activity}^e$. And $\mathcal{L}_{reg}$ is a regularization term that stabilizes the distribution of the edited vector by constraining it with Kullback-Leibler divergence.

Generator

The generator's task is to synthesize ECG signals by integrating a quantized health condition feature $\hat{f}_{health}^d$ with an edited activity level feature $f_{activity}^{d*}$. To achieve this objective, we modified the StyleGAN architecture [49]. Specifically, the activity level feature is transformed into a style vector s via a sequence of five linear layers, which is then injected into the health condition feature. For the injection process, we employ five sequential blocks for upsampling. In each block, the health condition feature $\hat{f}_{health}$ first undergoes a 1D convolution, followed by the addition of learnable noise. Unlike StyleGAN's approach, which directly fuses $\hat{f}_{health}$ and s using adaptive instance normalization (AdaIN) [50], we propose a position-aware instance normalization (PosIN) to integrate the features more effectively. In our case, the injection of activity style should vary across the ECG signal rather than being applied uniformly. For instance, the activity style predominantly influence the waveform's dynamic segments, such as the QRS complex, while leaving more stable segments like the T wave relatively unchanged. Given $\hat{f}_{health} \in \mathbb{R}^{C_h \times L_h}$ and $s \in \mathbb{R}^{L_s}$, they are transformed into the same shape by:

$$\begin{aligned}\hat{f}_{health} &= \phi(IN(W_f \hat{f}_{health} + b_f)), \hat{f}_{health} \in \mathbb{R}^{C' \times L'}, \\ s &= \text{reshape}(\phi(W_s s + b_s), (C', L')) + Pos, s \in \mathbb{R}^{C' \times L'}.\end{aligned}\quad (7)$$

where W_f , W_s , b_f and b_s are the weights and bias of the projection, Pos is a learnable positional embedding, $\phi(\cdot)$ denotes the LeakyReLU activation, and $\text{reshape}(\cdot, (C', L'))$ reshapes the output into shape $\mathbb{R}^{C' \times L'}$. We then perform an attention score a between $\hat{f}_{health}$ and s as $\mathcal{S}((W_a(\hat{f}_{health} \circ s) + b_a))$ where W_a and b_a are learnable parameters, $\circ$ represents concatenation along the channel dimension, and $\mathcal{S}(\cdot)$ denotes the sigmoid function. Finally, the normalization parameters μ and σ are computed by applying a linear transformation to the weighted combination $a \cdot \hat{f}_{health} + (1 - a) \cdot s$.

We design the generator's objective to consist of two components: an adversarial loss and a reconstruction loss. The adversarial loss follows the standard GAN formulation:

$$\mathcal{L}_{adv}^G = \mathbb{E}_{f, s \sim p_{\text{fake}}} [\log(1 - D(G(f, s)))], \quad (8)$$

where an ideal generator G produces fake ECGs that the discriminator D recognizes as real, outputting a value of 1. To encourage the generator to capture the fundamental wave and

peak characteristics of ECG signals efficiently, we incorporate a reconstruction loss. Specifically, we generate ECG signals using the health and activity features from abnormal ECGs, and then compute the difference between the generated and the real signals, presented as:

$$\mathcal{L}_{rec} = \|X_d - G(f_{activity}^d, f_{health}^d)\|^2. \quad (9)$$

Discriminator

Inspired by [32], our discriminator comprises two branches—one for time-domain and another for frequency-domain—to better preserve the informative content in the generated ECG signals. The time branch consists of five blocks, with each block comprising two 1D convolution layers, one ReLU activation, and one batch normalization layer. To obtain frequency-domain signals, we perform a Short-Time Fourier Transform (STFT) on the time series, defined as:

$$X(m, \omega) = \sum_{n=-\infty}^{\infty} x[n] w[n - m] e^{-j\omega n}, \quad (10)$$

where $x[n]$ denotes the time-domain signal, $w[n]$ is a window function, and m represents the hop length. The 2D spectrogram is then obtained by taking the logarithm of the magnitude of $X(m, \omega)$, i.e., $S(m, \omega) = \log(|X(m, \omega)|)$. The frequency branch also consists of five blocks similar to the time branch, while the 1D convolutions are replaced by 2D convolutions to process the spectrogram. For both the time and frequency branches, an additional average pooling layer and a linear layer are appended to finally determine the authenticity of the ECG. Therefore, the discriminator is trained by maximizing the following objective:

$$\mathcal{L}_{adv}^D = \mathbb{E}_{x \sim p_{\text{real}}} [\log D(x)] + \mathbb{E}_{\hat{x} \sim p_{\text{fake}}} [\log(1 - D(\hat{x}))], \quad (11)$$

where a perfect discriminator outputs 1 for ECGs from real data and 0 for ECGs from generated data.

Overall Training Process

The overall training process is divided into three steps as follows:

- Firstly, we update the parameters of S based on $\mathcal{L}_{clf}$, $\mathcal{L}_{vq}$, $\mathcal{L}_{edit}$ and $\mathcal{L}_{reg}$.
- Next, we fix the parameters of S and G , and update the parameters of D based on Equation 11.
- Finally, we compute the forward pass, and update the parameters of D with the objective function $\mathcal{L}_{rec} + \mathcal{L}_{adv}^G$.

EXPERIMENTS

In this section, we first validate the effectiveness of our approach through abnormal ECG classification. We then conduct a clinical Turing test to assess the fidelity of the synthesized ECGs. Furthermore, we compare real and synthetic ECGs by visualizing the distribution differences across 11 waveform features. Next, we perform ablation studies to evaluate the contribution of each model component. To further assess the effectiveness of feature disentanglement, we visualize the 2D t-SNE

Dataset	Length (Sec)	Frequency (Hz)	#Subjects	#Classes	#Leads	Usage
ECGID	400	300	45	2	1	train synthesizer (rest & exercise ECGs)
PTB-XL	10	500	18,885	5	12	train synthesizer (abnormal ECGs) & classifier
MuseECG	10	500	10,646	4	12	train synthesizer (abnormal ECGs) & classifier
MedalCare-XL	10	500	n.a.	8	12	train synthesizer (abnormal ECGs) & classifier
CinC 2017	30	300	8,528	3	1	train synthesizer (abnormal ECGs) & classifier
PTB	10	1000	290	5	12	cross-dataset validation

TABLE II: Statistics of datasets utilized.

Methods	PTB-XL Dataset					MuseECG Dataset				
	Accuracy	F1 score	AUROC	MCC	Kappa	Accuracy	F1 score	AUROC	MCC	Kappa
w/o augmentation	52.9 ± 1.4	36.6 ± 1.7	67.3 ± 1.2	29.1 ± 1.5	28.9 ± 1.5	92.3 ± 0.1	92.5 ± 0.1	98.1 ± 0.1	89.5 ± 0.2	89.5 ± 0.2
CGAN	56.1 ± 1.2	36.6 ± 1.2	67.6 ± 1.1	32.1 ± 1.4	31.4 ± 1.1	92.6 ± 0.2	92.7 ± 0.2	98.4 ± 0.2	89.9 ± 0.3	89.9 ± 0.3
PGAN	56.7 ± 1.3	36.7 ± 1.6	68.6 ± 1.6	33.0 ± 1.1	31.9 ± 1.7	92.7 ± 0.3	92.8 ± 0.3	98.3 ± 0.1	90.0 ± 0.4	90.0 ± 0.4
WGAN	56.3 ± 0.8	36.9 ± 1.2	68.9 ± 1.8	32.1 ± 1.0	31.4 ± 1.3	92.7 ± 0.2	92.8 ± 0.2	98.5 ± 0.2	90.1 ± 0.3	90.1 ± 0.3
StyleGAN	56.4 ± 0.8	34.2 ± 1.6	68.4 ± 1.2	31.3 ± 1.4	30.1 ± 1.2	92.7 ± 0.3	92.9 ± 0.2	98.7 ± 0.1	90.1 ± 0.3	90.1 ± 0.3
SimGAN	55.9 ± 1.2	36.7 ± 1.6	68.3 ± 1.2	32.1 ± 1.2	31.6 ± 1.1	92.8 ± 0.2	93.0 ± 0.2	98.5 ± 0.2	<u>90.2</u> ± 0.3	<u>90.2</u> ± 0.3
CardioGAN	57.1 ± 1.2	36.6 ± 1.9	67.2 ± 1.4	33.1 ± 1.4	32.1 ± 1.1	92.4 ± 0.3	92.5 ± 0.3	98.4 ± 0.1	89.7 ± 0.4	89.7 ± 0.4
Diffusion-SSSM	57.0 ± 1.4	<u>37.2</u> ± 1.6	67.5 ± 1.4	<u>33.2</u> ± 1.6	<u>32.4</u> ± 1.5	92.1 ± 0.3	92.3 ± 0.4	98.4 ± 0.3	<u>90.2</u> ± 0.3	<u>90.2</u> ± 0.3
ECG twins	53.8 ± 0.9	34.0 ± 1.8	67.9 ± 1.4	28.4 ± 1.2	27.9 ± 1.1	91.6 ± 0.4	91.6 ± 0.4	98.2 ± 0.2	88.2 ± 0.5	88.2 ± 0.5
Ours	58.6* ± 1.0	37.6** ± 1.4	70.0* ± 0.8	34.6** ± 1.5	33.5** ± 1.0	93.7** ± 0.2	93.6** ± 0.3	98.8 ± 0.0	91.7** ± 0.4	91.7** ± 0.4
Methods	MedalCare-XL Dataset					CinC 2017 Dataset				
	Accuracy	F1 score	AUROC	MCC	Kappa	Accuracy	F1 score	AUROC	MCC	Kappa
w/o augmentation	93.9 ± 0.5	88.8 ± 1.1	99.6 ± 0.1	91.2 ± 0.8	91.1 ± 0.8	74.3 ± 0.5	68.8 ± 0.8	83.6 ± 0.6	53.8 ± 1.0	53.6 ± 1.0
CGAN	94.3 ± 0.7	89.3 ± 1.6	99.7 ± 0.1	91.8 ± 1.0	91.7 ± 1.0	74.9 ± 0.6	68.5 ± 1.0	84.4 ± 0.9	52.4 ± 1.3	52.0 ± 1.2
PGAN	94.3 ± 0.5	89.2 ± 1.1	99.7 ± 0.1	91.7 ± 0.7	91.7 ± 0.7	76.1 ± 0.8	70.1 ± 1.5	85.1 ± 0.8	<u>55.1</u> ± 1.5	54.8 ± 1.4
WGAN	94.3 ± 0.6	89.1 ± 1.4	99.6 ± 0.1	91.8 ± 0.8	91.8 ± 0.9	75.6 ± 0.4	69.7 ± 0.6	85.1 ± 0.5	53.6 ± 0.8	53.3 ± 0.8
StyleGAN	94.4 ± 0.6	89.4 ± 1.3	99.7 ± 0.0	91.9 ± 0.9	91.9 ± 0.9	<u>76.1</u> ± 1.1	<u>70.4</u> ± 1.1	84.9 ± 0.7	54.7 ± 1.9	54.2 ± 1.6
SimGAN	94.5 ± 0.8	90.0 ± 1.6	99.6 ± 0.0	92.5 ± 0.5	92.4 ± 0.5	75.7 ± 0.8	69.9 ± 1.2	85.2 ± 0.7	54.0 ± 1.5	53.7 ± 1.5
CardioGAN	94.7 ± 0.3	90.5 ± 0.5	99.7 ± 0.0	92.4 ± 0.0	92.4 ± 0.0	74.5 ± 0.7	67.6 ± 1.1	83.6 ± 0.5	51.1 ± 1.6	50.5 ± 0.6
Diffusion-SSSM	<u>95.8</u> ± 0.5	<u>92.5</u> ± 1.1	99.8 ± 0.0	<u>93.7</u> ± 0.4	<u>93.7</u> ± 0.4	74.0 ± 1.0	67.4 ± 1.3	83.0 ± 0.9	50.6 ± 2.0	50.5 ± 2.0
ECG twins	90.5 ± 0.5	82.3 ± 1.4	98.9 ± 0.3	86.4 ± 0.9	86.3 ± 0.9	74.4 ± 0.5	68.3 ± 0.5	83.7 ± 0.8	51.7 ± 0.9	51.4 ± 0.9
Ours	96.1 ± 0.6	93.8** ± 0.8	<u>99.7</u> ± 0.0	94.3 ± 0.8	94.3 ± 0.8	76.9** ± 0.4	72.0** ± 0.9	85.9* ± 0.4	56.0** ± 0.8	55.6** ± 0.9

TABLE III: Comparison with state-of-the-art baselines on abnormal ECG classification. We use **bold** to indicate the best results and underline for the second best one. The second-best method is determined based on AUROC. Statistical significance is denoted as * for $p < 0.05$ and ** for $p < 0.01$ when compared with the second-best method.

Methods	PTB Dataset				
	Accuracy	F1 score	AUROC	MCC	Kappa
w/o augmentation	41.6 ± 4.7	24.1 ± 1.5	61.7 ± 2.0	14.7 ± 1.6	12.1 ± 1.7
CGAN	46.4 ± 7.1	26.2 ± 2.1	<u>64.3</u> ± 3.1	19.7 ± 2.6	<u>16.3</u> ± 2.7
PGAN	45.3 ± 8.1	25.3 ± 2.6	62.0 ± 2.4	17.7 ± 2.2	14.5 ± 2.0
WGAN	45.5 ± 4.6	25.6 ± 2.0	63.7 ± 2.8	18.8 ± 3.7	15.3 ± 2.9
StyleGAN	43.2 ± 6.2	26.6 ± 2.3	62.3 ± 1.2	<u>20.2</u> ± 2.1	<u>16.3</u> ± 2.8
SimGAN	45.2 ± 7.0	25.5 ± 2.1	61.5 ± 3.0	17.2 ± 4.6	14.3 ± 4.1
CardioGAN	42.0 ± 4.4	23.9 ± 1.8	63.0 ± 2.4	19.1 ± 1.8	15.0 ± 2.0
Diffusion-SSSM	44.9 ± 2.9	<u>26.6</u> ± 2.3	62.0 ± 2.3	18.6 ± 2.5	15.4 ± 2.0
ECG Twins	40.4 ± 3.0	24.7 ± 1.9	62.8 ± 2.8	16.6 ± 2.5	13.0 ± 1.8
Ours	47.9 ± 2.4	27.7* ± 1.5	67.0 ± 2.4	21.8* ± 1.4	18.3* ± 1.0

TABLE IV: Cross-dataset validation on the PTB dataset. We use **bold** to indicate the best results and underline for the second best one. The second-best method is determined based on AUROC. Statistical significance is denoted as * for $p < 0.05$ and ** for $p < 0.01$ when compared with the second-best method.

embeddings of the original ECGs, the health condition features, and the activity level features. We also demonstrate the impact of activity feature injection using Grad-CAM visualizations. In addition, we discuss the effects of preprocessing and noise interference, highlighting the robustness of our approach. Finally, we report the model complexity in terms of parameter count, training time, and inference speed. Due to space limitations, we report only the most essential results below. Additional experimental results can be found in the supplementary materials at: <https://github.com/XiaoShuhong/GAEA-NET/blob/main/appendix.pdf>

Performance of Abnormal ECG Classification

Datasets: Table II presents all the datasets involved in this evaluation. The ECGID dataset [38] consists of single-lead ECG recordings collected from 45 healthy subjects during structured exercise sessions. The data includes both rest and exercise segments, serves as the activity level reference required to train our synthesizer. Five abnormal ECG datasets were included. Among them, MuseECG [51], MedalCare-XL [17], and CinC 2017 [19] primarily focus on arrhythmia-related conditions, such as atrial fibrillation, while PTB-XL [10], [52] and PTB [20] address other cardiovascular con-

ditions, including myocardial infarction and ST/T changes. Considering the limited number of samples in the PTB dataset, it was used only for cross-dataset validation. Specifically, the classifier trained on PTB-XL was tested on PTB without further fine-tuning. The other four abnormal datasets were partitioned into training and testing sets following previous studies [14], [53], ensuring that data from the same subject did not appear in both sets. These training sets were used for synthetic model, with generated data subsequently incorporated as augmented samples during classifier training. The classifier's performance was then evaluated on the test set. Further details of the datasets can be found in Appendix A.

To simulate real-world personal device scenarios, we exclusively utilized Lead I data across all datasets. We applied a consistent preprocessing procedure for all raw ECG data. A low-pass Butterworth filter [54] was used to eliminate electromyographic (EMG) noise above 40 Hz. ECG peak information was obtained using the NeuroKit2 package. Based on the locations of R-peaks, the original recordings were segmented into sequences containing seven heartbeats, with an additional 0.15 seconds retained before the first R-peak and after the last R-peak of each segment. To ensure consistency across datasets, these segments were uniformly resampled at 300 Hz and scaled proportionally to a fixed length of 1024 points.

Baseline Methods and Metrics: We compare our approach against eight state-of-the-art methods, including CGAN [55], PGAN [4], WGAN [56], StyleGAN [49], SimGAN [12], CardioGAN [13], Diffusion-SSSM [30], and ECG Twins [14]. A description of these baselines can be found in Appendix B. We employed a 1D ResNet-34 [21] as the standard classifier. For evaluation metrics, we used accuracy, F1-score, the area under the receiver operating characteristic curve (AUROC), Matthews correlation coefficient (MCC) and Cohen's Kappa, which are widely adopted for assessing ECG classification performance. To ensure a fair evaluation, we assessed the performance 10 times across consecutive seeds and reported the mean and standard deviation.

Implementation and Training: For our feature separator, we set both the activity level and health condition feature dimensions to $\in \mathbb{R}^{512 \times 32}$ after encoding. In the quantization module, N is set to 128 learnable vectors, each with a dimension of 512. In the offset editor, the activity latent space z has a dimension of 128, and we employ 8 heads in the multi-head cross attention mechanism. For our generator, all 1D convolutions use a kernel size of 3 and a stride of 1. The style vector s is set to a dimension of 1024, and the learnable position embedding Pos in our PosIN module is initialized to zero. Regarding the discriminator, a Hann window is used in the STFT, with a hop length of 8. Both 1D and 2D convolution layers also adopt a kernel size of 3 and stride of 1. Finally, all dropout layers are set to 0.2.

All experiments were performed on a server with NVIDIA GeForce RTX 3090 24GB and PyTorch 2.4.0+cu124. For synthesizer training, we used a batch size of 128, a uniform learning rate of $1e-4$ for all components, and an EMA decay of 0.99. A total of 50,000 iterations required roughly 15 hours on a single GPU. For classifier training, each method

generated approximately 10,000 synthesized samples per class to augment the training set. We set the batch size to 128, used random seeds from 777 to 786 for 10 trials, and trained for 50 epochs per trial. The initial learning rate was $1e-4$, decaying by a factor of 10 at the 20^{th} and 40^{th} epochs. Each trial took about 1 hour.

Results: As a clinically measured exercise-induced abnormal ECG dataset is not available, our classification evaluation was conducted only on resting abnormal datasets. Nevertheless, even in the resting state, individual differences in metabolism and physiology can lead to varying baseline activity levels. Our method effectively addresses these inter-individual discrepancies, resulting in improved classification performance. This improvement demonstrates the potential of our approach for addressing more complex classification during exercise scenarios. As shown in Table III, our method consistently outperformed all baseline approaches across the four datasets.

On **PTB-XL**, our model demonstrated clear advantages over the second-best WGAN, with improvements observed across all five metrics. Specifically, we achieved an accuracy of 58.6%, F1 score of 37.6%, AUROC of 70.0%, MCC of 34.6%, and Kappa of 33.5%. The gains were statistically significant: accuracy (+2.3%, $t(9) = 3.34$, $p = 0.01^*$), F1 score (+0.7%, $t(9) = 3.32$, $p < 0.01^{**}$), AUROC (+1.1%, $t(9) = 3.18$, $p = 0.01^*$), MCC (+2.5%, $t(9) = 4.19$, $p < 0.01^{**}$), and Kappa (+2.1%, $t(9) = 5.16$, $p < 0.01^{**}$).

MuseECG results also highlight the superiority of our approach. We obtained 93.7% accuracy and 93.6% F1 score, significantly outperforming StyleGAN by +1.0% in accuracy ($t(9) = 10.39$, $p = 0.00^{**}$) and +0.7% in F1 ($t(9) = 5.88$, $p = 0.00^{**}$). Notably, improvements in MCC and Kappa were also significant, while AUROC showed a marginal, non-significant increase (+0.1%, $p = 0.71$).

For **MedalCare-XL**, the overall performance was strong, with 96.1% accuracy, 93.8% F1, and 99.7% AUROC. In comparison with Diffusion-SSSM, the F1 score showed a statistically significant improvement (+1.3%, $t(9) = 5.01$, $p < 0.01^{**}$), while the remaining metrics showed comparable results without significance.

Regarding the **CinC 2017**, our model again achieved superior results—76.9% accuracy, 72.0% F1 score, 85.9% AUROC, 56.0% MCC, and 55.6% Kappa. Compared with SimGAN, statistically significant improvements were observed across the board, such as +2.1% in F1 score ($t(9) = 5.43$, $p < 0.01^{**}$) and +2.0% in MCC ($t(9) = 3.80$, $p < 0.01^{**}$).

In addition, Appendix C provides class-wise ROC curves and the corresponding AUC scores with 95% confidence intervals, further confirming the robustness of our approach across different diagnostic categories.

Table IV reports the results of the cross-dataset generalization evaluation on **PTB**. When testing a PTB-XL-trained classifier on PTB data, our method outperformed CGAN in three of five metrics: F1 score (+1.5%, $t(9) = 2.44$, $p = 0.04^*$), MCC (+2.1%, $t(9) = 3.12$, $p = 0.01^*$), and Kappa (+2.0%, $t(9) = 2.62$, $p = 0.03^*$). Although improvements were also observed in accuracy and AUROC, they were not statistically significant. These results suggest that our method generalizes

ID	Gender	Age	Professional Title	Time on Task 1	Time on Task 2
P1	F	37	Attending Physician	25	39
P2	F	32	Resident Physician	38	41
P3	F	37	Resident Physician	15	40
P4	M	33	Resident Physician	27	42
P5	M	29	Resident Physician	40	80
P6	M	42	Attending Physician	23	43
P7	F	34	Resident Physician	60	30

TABLE V: Statistics of Experts involved.

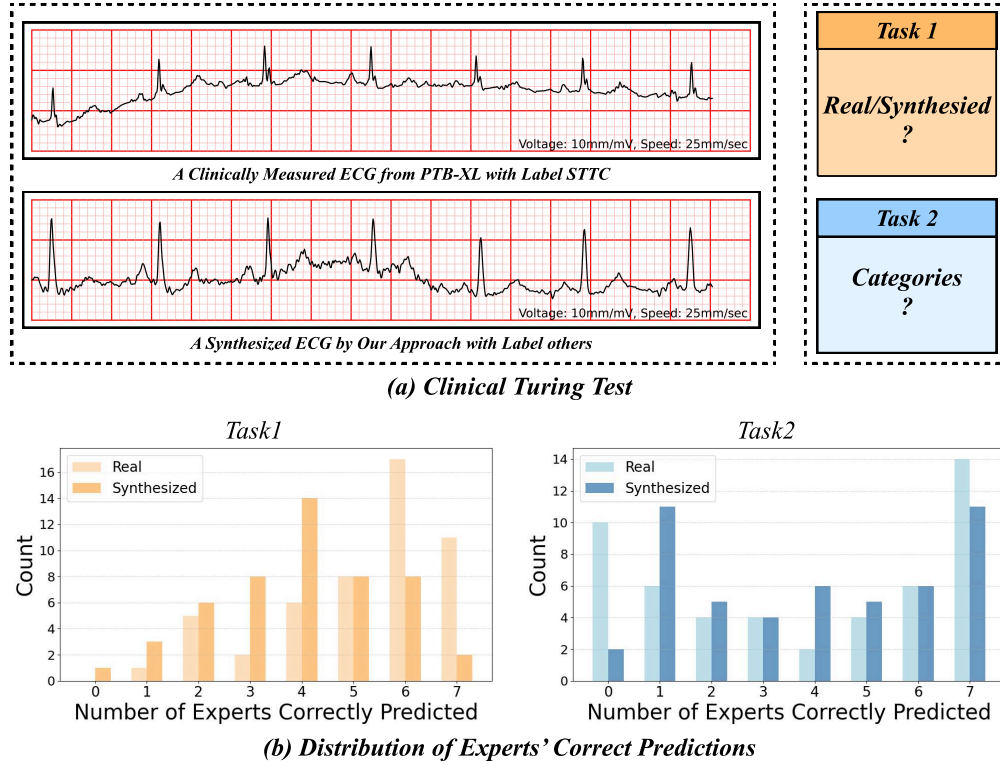


Fig. 2: Clinical Turing test and results. (a) Illustrating the clinical Turing test procedure, where ECG samples are displayed on standard ECG chart paper with grids. (b) Showing the distribution of experts' correct predictions, where the x-axis indicates the number of experts (from 0 to 7), and the y-axis shows the corresponding count of ECG signals.

well to unseen datasets without retraining.

Clinical Turing Test

Procedure: Similar to [57], we aim to ensure that the synthetic ECG signals correspond to clinically measured signals rather than merely reflecting statistical artifacts. To this end, we invited 7 experienced cardiologists to participate in a clinical Turing test, with their information presented in Table V. Specifically, each cardiologist was assigned 100 ECGs, including 50 clinically measured samples from the PTB-XL dataset and their corresponding synthetic samples generated by our method. Each sample was presented on a separate standard ECG chart paper, as shown in Figure 2(a), where each small grid represents 1 mm. The time and voltage scales were annotated in the lower right corner to facilitate evaluation.

During the test, the cardiologists assessed each ECG and perform two tasks. First, they were asked to determine whether

each ECG was real or synthetic. They were not informed of the proportion of real and synthetic samples in advance, leaving them to just rely on their professional judgment. Moreover, the fact that the synthetic ECGs were designed to simulate exercise-induced patterns was withheld to prevent introducing additional classification bias. Second, experts were required to provide diagnostic classifications for each ECG. The PTB-XL dataset includes five categories—normal ECG, myocardial infarction, ST/T changes, conduction disturbance, and hypertrophy. However, based on discussions with cardiologists prior to the formal evaluation, the following revision to the original category descriptions was provided to the experts:

- The hypertrophy category was excluded, as it is generally difficult to diagnose when only lead-I data are available [11].
- Multiple categories could be assigned to a single ECG signal, particularly because ST/T changes often co-occur

Experts	# Real	# Synthetic	Accuracy	Precision	Recall	F1 score
P1	65	35	61.0	58.5	76.0	66.1
P2	78	22	58.0	55.1	86.0	67.2
P3	62	38	74.0	69.3	86.0	76.8
P4	49	51	69.0	69.4	68.0	68.7
P5	60	40	70.0	66.7	80.0	72.7
P6	33	67	59.0	63.6	42.0	50.6
P7	66	34	66.0	62.1	82.0	70.7
–	59	41	65.3 [60.8-69.8]	63.5 [59.5-67.5]	74.3 [62.8-85.8]	67.5 [61.4-73.7]

TABLE VI: Performance on Task 1. We report the number of samples identified as real and synthetic by each expert, along with their corresponding accuracy, precision, recall, and F1 score. The last row indicates the average performance across all experts. Values in square brackets denote the 95% confidence interval.

Experts	Real ECG				Synthetic ECG			
	Accuracy	Precision	Recall	F1 score	Accuracy	Precision	Recall	F1 score
P1	64.0	76.1	57.9	60.7	62.0	77.9	55.2	55.7
P2	52.0	79.7	43.8	44.5	50.0	60.6	41.8	41.3
P3	58.0	80.6	51.3	54.0	50.0	72.9	42.0	42.6
P4	54.0	60.1	48.5	48.5	62.0	77.9	55.5	58.4
P5	50.0	57.8	45.7	45.3	52.0	59.0	46.1	45.4
P6	52.0	57.7	44.2	42.2	48.0	49.9	40.4	37.1
P7	54.0	68.1	46.9	47.2	66.0	70.5	64.6	66.4
–	54.9 [51.3-58.4]	68.6 [61.0-76.2]	48.3 [44.7-52.0]	48.9 [44.2-53.7]	55.7 [50.3-61.2]	67.0 [59.1-74.9]	49.4 [42.6-56.2]	49.6 [41.6-57.5]

TABLE VII: Performance on Task 2. We compare experts’ diagnostic performance between real and synthetic ECG signals, reporting accuracy, precision, recall, and F1 score. The last row indicates the average performance across all experts. Values in square brackets denote the 95% confidence interval.

with other abnormalities such as myocardial infarction.

- An “others” category was introduced to replace the normal ECG category and also to encompass conditions not defined in PTB-XL (e.g., premature atrial contractions, premature ventricular contractions, supraventricular tachycardia) when identified.

Results: As shown in Table V, 7 cardiologists (P1-P7, 4 female and 3 male) aged 29–42 ($M = 34.86$, $SD = 4.22$) participated in our evaluation. For Task 1, the mean completion time was 32 minutes ($SD = 15$), whereas Task 2 took an average of 45 minutes ($SD = 16$). Figure 2(b) shows the distribution of correct predictions made by the experts. For example, the columns corresponding to “7” in Task 1 indicates that 11 real data samples were correctly identified by all 7 experts, while only 2 synthetic signals were consistently identified as fake.

The primary objective of Task 1 was to evaluate whether the experts were able to reliably distinguish between real and synthetic ECG signals. First, we assessed inter-rater consistency among the seven cardiologists. The resulting Kappa score of 0.14 reflects only slight agreement, indicating that distinguishing synthetic signals from real ones is challenging even for experienced cardiologists. Table VI further presents the statistical results for Task 1, where we recorded the number of samples that experts identified as real or synthetic and evaluated metrics including accuracy, precision, recall, and F1 score. Overall, the experts achieved an average accuracy of 65.3% (95% CI: 60.8-69.8), a precision of 63.5% (95% CI: 59.5-67.5), a recall of 74.3% (95% CI: 62.8-85.8), and an F1 score of 67.5% (95% CI: 61.4-73.7). Specifically, the recall scores indicate that the cardiologists retained a certain ability to identify synthetic signals. According to their feedback, this

was largely because the activity-enhanced synthetic ECGs tended to exhibit a more unstable baseline than real ones obtained under resting conditions, making some synthetic signals easier to spot.

In Task 2, the experts were asked to assign diagnostic categories to each ECG. By comparing their diagnostic performance separately on real and synthetic recordings, we aimed to verify whether our method preserved the pathological cues essential for diagnosis—that is, cardiologists could still deliver consistent interpretations when accessing the synthetic ECGs. A paired t-test was conducted to compare diagnostic performance on real versus synthetic ECGs. Statistically, no significant differences were detected across any metric: accuracy ($t(6) = -0.32, p = 0.76$), precision ($t(6) = 0.38, p = 0.72$), recall ($t(6) = -0.31, p = 0.77$), and F1 score ($t(6) = -0.16, p = 0.88$), and the numerical values for each measure were nearly identical between the two groups. In addition, an inter-group consistency analysis yielded a Kappa coefficient of 0.48, indicating moderate agreement. These findings indicate that the synthetic ECGs preserve the abnormal feature patterns found in real data, thereby supporting experts in making consistent diagnoses. Finally, we also noted that the experts’ overall accuracy in this test was lower than their clinical performance. This is mainly attributable to the fact that, in clinical practice, cardiologists work with ECGs of substantially longer duration and higher dimensions, which allows for more thorough verification than the shorter, simplified signals supplied in our experiment, as noted by the participating experts.

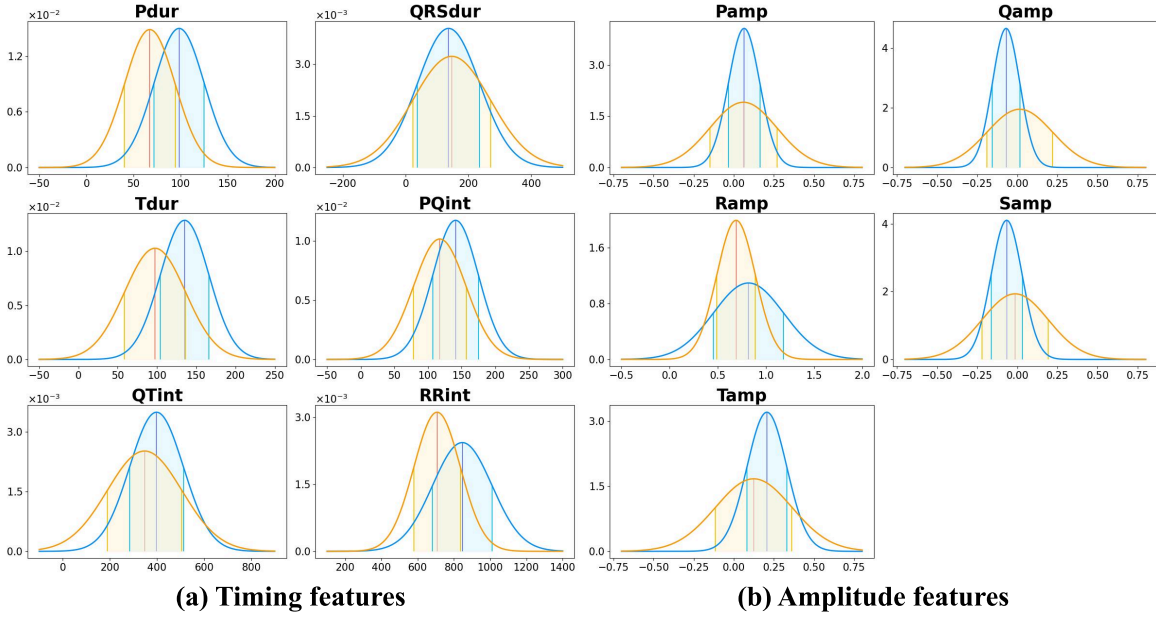


Fig. 3: Comparison of features in the clinical and synthetic cohorts. Probability density functions are shown for (a) 6 timing features and (b) 5 amplitude features. The yellow and blue curves represent synthetic and real signal distributions, respectively. Centered vertical lines highlight the mean values μ , and the shaded areas represent one standard deviation intervals $[\mu - \sigma, \mu + \sigma]$.

Connection	Quantization	Offset Editor	PosIN	$\mathcal{L}_{frequency}$	Accuracy	F1 score	AUROC
✓	✓	✓	✓	✓	58.7 ± 1.1	37.6 ± 1.5	69.8 ± 0.6
×	✓	✓	✓	✓	57.9 ± 1.5	35.5 ± 1.5	67.3 ± 1.8
✓	×	✓	✓	✓	55.6 ± 1.0	36.9 ± 0.8	68.2 ± 1.5
✓	✓	×	✓	✓	56.1 ± 1.2	35.9 ± 1.4	69.0 ± 1.6
✓	✓	✓	×	✓	57.3 ± 1.1	34.6 ± 1.6	68.0 ± 1.5
✓	✓	✓	✓	×	57.1 ± 0.7	35.2 ± 1.4	68.4 ± 1.1
×	×	×	×	×	53.1 ± 1.9	33.2 ± 1.7	66.9 ± 1.1

TABLE VIII: Ablation Studies.

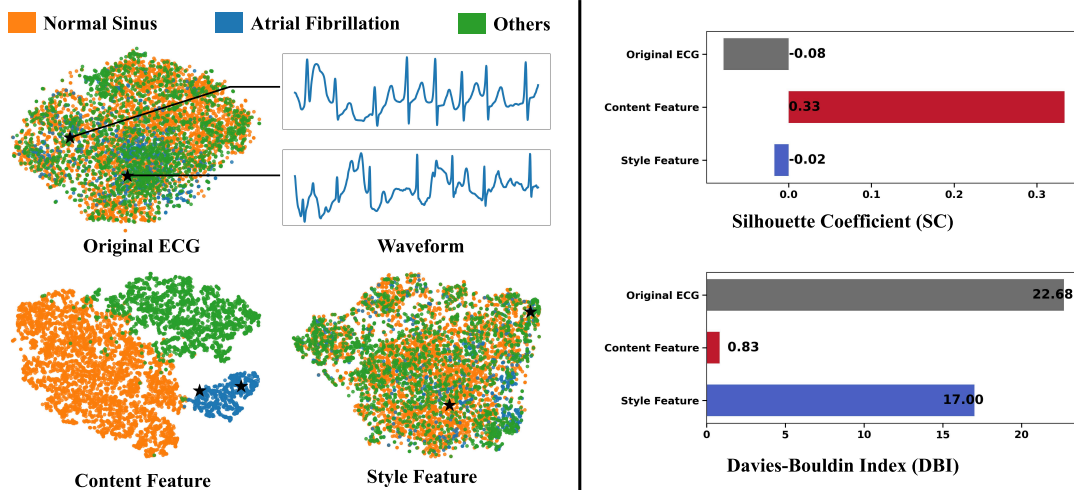


Fig. 4: Visualization of feature separation. We present the 2D t-SNE embeddings of the original ECG signals, style features, and content features. The clustering performance is reported using the Silhouette Coefficient (SC) and Davies-Bouldin Index (DBI).

Feature Distribution

Procedure: To further compare the statistical properties of the synthetic and real data, we calculated 11 handcrafted

features commonly used to characterize ECG signals [17]. These include six time-domain features: P wave duration, QRS duration, T wave duration, PQ interval, QT interval, and RR

interval. And five amplitude features: P wave amplitude, Q peak amplitude, R peak amplitude, S peak amplitude, and T wave amplitude. This experiment was conducted on the PTB-XL dataset, where for each real sample, our method generated a corresponding synthetic sample. All the necessary feature information was extracted using the NeuroKit2 package.

Results: Figure 3 presents the probability density functions for the six time-domain and five amplitude features. The yellow and blue curve showcase the distribution of the synthetic and real clinical data. The centered vertical lines highlight the mean value μ , and the shaded area represents one standard deviation, i.e., the range $[\mu - \sigma, \mu + \sigma]$. For all 11 features, real and synthetic data showed a high degree of coincide only in QRS duration (overlap = 0.87). In contrast, the synthetic features covered only a subset of the clinically observed range, with overlap values of 0.49 for P wave duration, 0.50 for T wave duration, 0.64 for PQ interval, 0.52 for QT interval, 0.54 for RR interval, 0.55 for P wave amplitude, 0.50 for Q peak amplitude, 0.69 for R peak amplitude, 0.51 for S peak amplitude, and 0.55 for T wave amplitude. Overall, these results align with previous findings [17] and the ranges of all these features are still considered clinically reasonable.

Ablation Studies

Table VIII presents the results of our ablation studies. Overall, we validated the effectiveness of five key components, which together yielded improvements of 5.6% in accuracy, 4.4% in F1 score, and 2.9% in AUROC. In terms of individual components, the “Connection” module—which introduces inter-encoder information exchange within the separator—resulted in an improvement of 0.8% in accuracy, 2.1% in F1 score, and 2.5% in AUROC. The “Quantization” module, designed to further remove subject-specific variability from the health condition features, contributed an enhancement of 3.1% in accuracy, 0.7% in F1 score, and 1.6% in AUROC. For the “Offset Editor”, our approach outperformed the direct use of activity-level features from exercise ECGs, achieving an increase of accuracy, F1 score and AUROC by 2.6%, 1.7%, and 0.8%. Additionally, compare to the conventional AdaIN, the position-aware design of our “PosIN” module facilitated the dynamic injection of activity-level features based on position, leading to a gain of 1.4% in accuracy, 3.0% in F1 score, and 1.8% in AUROC. Finally, the frequency branch of the discriminator “ $\mathcal{L}_{frequency}$ ” delivered an improvement of 1.6% in accuracy, 2.4% in F1 score, and 1.4% in AUROC, demonstrating that control over frequency-domain information provides additional benefits beyond those achieved through time-domain.

Visualization of Feature Separation

Due to individual-specific variations, ECG signals within the same abnormal category can exhibit considerable differences. To address this issue and bring samples with the same abnormal label closer together, we introduce a feature separator that encodes the ECG signal into disease domain-invariant activity level features (style) and domain-specific

health condition features (content). In this section, we validated the effectiveness of this strategy by visualizing the 2D distribution of the original ECG signals, style features, and content features using t-SNE. Here, we present the results on the CinC 2017 dataset, which includes three primary rhythm categories: normal sinus rhythm, atrial fibrillation, and other rhythms. Additional results can be found in Appendix D. The Silhouette Coefficient (SC) and Davies-Bouldin Index (DBI) were utilized to evaluate the clustering performance of the features. SC measures the cohesion within each category, where a larger value indicates better concentration. On the other hand, DBI focuses on the ratio of inter-category to intra-category distance, where a smaller value indicates better concentration quality.

Figure 4 presents the visualization results. Both the original ECGs (SC=-0.08, DBI=22.68) and style features (SC=-0.02, DBI=17.04) are mixed, whereas the content features (SC=0.33, DBI=0.83) demonstrate clear distinguishability across the three categories. Furthermore, we randomly selected two ECGs belonging to atrial fibrillation and show their waveforms. Due to individual differences, they exhibit significant morphological variations, which result in a Euclidean distance of 18.67 between them. As the domain-invariant personal specificity is eliminated in the content features, the Euclidean distance between them decreases to 13.65 as we expected. And, the distance between these two samples in the style features increases, reaching 38.91, which we interpreted as the individual-specific characteristics.

While current results are promising, the effectiveness of our approach could be further validated in future clinical settings—for example, by comparing abnormal ECGs of a subject with their own historical exercise-related healthy ECGs recorded at different time points. Such comparisons could provide a more direct evaluation of the model’s ability to handle intra-subject variations under different physiological conditions.

Visualization of Activity Level

In this section, we evaluate the effectiveness of our activity-level feature injection. In particular, we trained a binary classifier based on ResNet-34 using paired rest and exercise ECGs to distinguish between different activity levels. This classifier was then applied to a test set containing real abnormal ECGs (representing low activity levels at rest) and synthetic abnormal ECGs (with high activity levels). The classifier achieved strong performance, with 87% accuracy, 93% precision, 88% recall, and 91% F1 score. Additionally, Grad-CAM [58] was employed to visualize the cues that the classifier relied on to differentiate activity levels. Specifically, the second convolution layer of the final ResNet block was used for gradient visualization. As shown in Figure 5, we present (a) a rest ECG with low activity level, (b) an exercise ECG with a high activity level, (c) an abnormal ECG from PTB-XL with low activity level, and (d) a synthetic abnormal ECG enriched with a high activity level. In these visualizations, the black line represents the original signal, and the blue portions highlight the classifier’s attention at

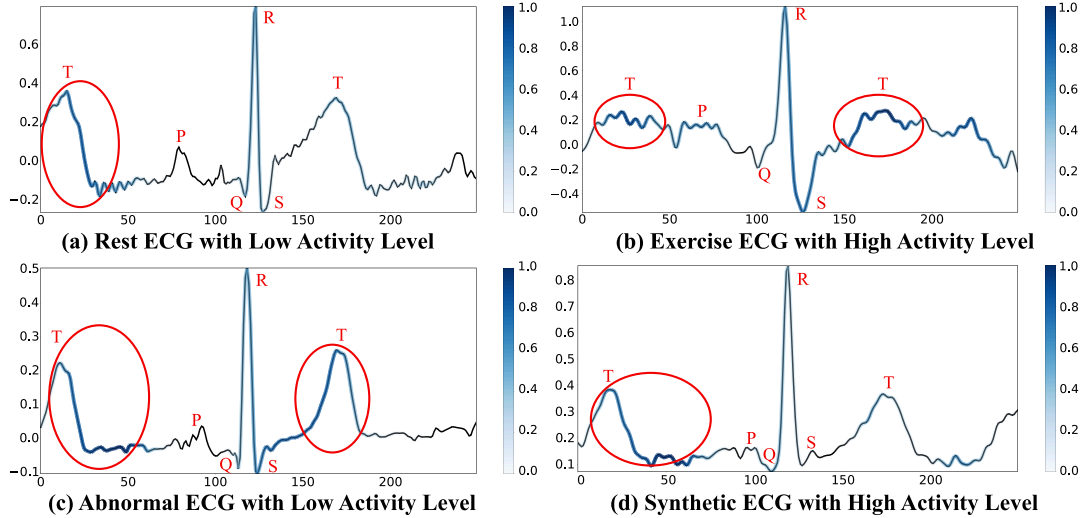


Fig. 5: Visualization of activity level through Grad-CAM. The black line represents the ECG signal, and the blue portions highlight the magnitude of the classifier’s gradients at each time point.

Category	Condition	Accuracy	F1 Score	AUROC	MCC	Kappa
Reference	Default	74.9 $\pm$ 0.5	70.3 $\pm$ 0.9	84.0 $\pm$ 0.6	53.1 $\pm$ 1.0	52.5 $\pm$ 0.8
Noise Interference	EMG Removal	75.5 $\pm$ 0.6	71.5 $\pm$ 1.3	84.7 $\pm$ 0.7	54.5 $\pm$ 1.0	54.2 $\pm$ 0.9
	Baseline Correction	77.0 $\pm$ 0.5	71.9 $\pm$ 0.8	85.4 $\pm$ 0.5	55.8 $\pm$ 0.7	55.7 $\pm$ 0.7
	AWGN (15dB)	74.9 $\pm$ 0.5	69.5 $\pm$ 1.2	83.7 $\pm$ 0.8	53.0 $\pm$ 1.0	52.4 $\pm$ 0.9
	AWGN (6dB)	73.2 $\pm$ 0.5	65.0 $\pm$ 1.0	82.8 $\pm$ 0.8	49.6 $\pm$ 0.7	49.2 $\pm$ 0.7
Singal Length	3 heartbeats	67.9 $\pm$ 0.8	58.3 $\pm$ 1.0	79.8 $\pm$ 0.6	39.2 $\pm$ 1.1	39.0 $\pm$ 1.0
	14 heartbeats	79.5 $\pm$ 0.5	75.6 $\pm$ 1.0	89.2 $\pm$ 0.4	63.4 $\pm$ 1.1	63.2 $\pm$ 1.1

TABLE IX: Robustness analysis

each time point. The darker the blue color, the higher the gradient, indicating that the model relies more on the signal features at these moments to make classification decisions. For instance, we observed higher Grad-CAM values around the T-wave, and the model also pays considerable attention to certain relatively flat ST segments or the later part of the T-wave. This suggests that the model uses features like ST-segment elevation and T-wave inversion to aid its decision-making, which is consistent with the principles used in clinical practice. Additionally, some activation is seen in the QRS complex, indicating that the QRS morphology changes under resting and exercise conditions. Overall, these observations indicate that our activity feature injection is effective, and our GAEGAN has successfully learned activity-enhanced ECGs that align with clinical knowledge.

Robustness Analysis

In this section, we performed a robustness analysis to evaluate the stability of our method under different signal processing conditions. The experiments were conducted on the CinC 2017 dataset. In practical scenarios, signals are often affected by noise interference. Therefore, we first examined different noise processing strategies. As a default setting, no denoising was applied to the raw signals. Under the EMG noise removal condition, we employed a low-pass Butterworth filter to suppress high-frequency components above 40Hz [54]. For baseline correction, we applied Locally Estimated

Scatterplot Smoothing (LOESS) to reduce the impact of low-frequency baseline wandering [17]. We also simulated scenarios with more severe noise interference by adding additive white Gaussian noise (AWGN) to the raw signals. Specifically, we evaluated two noise levels, where the signal-to-noise ratio (SNR) was set to 15 dB and 6 dB. The 15 dB condition represents moderate noise, while 6 dB corresponds to a severe noise level [59].

As shown in Table IX, the model achieved an accuracy of 74.9%, F1 score of 70.3%, AUROC of 84.0%, MCC of 53.1%, and Kappa of 52.5% under the default setting. When applying EMG noise removal, all metrics showed slight improvements, with F1 increasing to 71.5% and AUROC to 84.7%. Baseline correction led to the highest overall performance, with accuracy reaching 77.0%, F1 score 71.9%, AUROC 85.4%, MCC 55.8%, and Kappa 55.7%, indicating that mitigating low-frequency drift substantially benefits model stability. Under moderate noise (SNR=15 dB), our method demonstrated strong robustness, with only a minor drop of 0.7% in F1 score and 0.3% in AUROC compared to the default setting. When suffering severe noise conditions (SNR=6 dB), performance declined more noticeably, with both MCC and Kappa dropping below 50%.

Another factor we considered is the length of the ECG signal. Under the default setting, each ECG sample contains 7 heartbeats. Here, we evaluated the model’s performance using shorter (3 heartbeats) and longer (14 heartbeats) segments for

comparison. Overall, the classification performance improved with increasing signal length. The model achieved an accuracy of 67.9% with 3 heartbeats, 74.9% under the default setting with 7 heartbeats, and 79.5% with 14 heartbeats. This result aligns with expectations, as longer ECG segments provide more contextual information beneficial for both synthesis and classification.

Model Complexity

Methods	Params	Training Time	Inference Speed
CGAN	1.59	1	1807.91
PGAN	1.15	1	1702.13
WGAN	0.94	1	1815.60
StyleGAN	51.33	34	7.92
SimGAN	1.59	1.4	1718.12
CardioGAN	77.47	18	117.90
Diffusion-SSSM	0.80	1.5	5.61
ECG Twins	16.79	7	1177.55
Ours	18.01	15	563.13

TABLE X: Model complexity: Params (M), Training Time (h), Inference Speed (samples/s).

The computational complexity of all models is summarized in Table X. Specifically, we list each model’s parameter count, training time, and inference speed. The most compact networks, such as PGAN and Diffusion-SSSM, contain only about 1 M parameters, while the largest, CardioGAN, has 77.47 M parameters. Training times also show a wide spread, from 1 h to 34 h. Our model has 18.01 M parameters and requires roughly 15 h of training on a single NVIDIA GeForce RTX 3090 (24 GB). For inference speed, we measure it by the number of synthetic ECG samples generated per unit of time. In one second, our model can generate 563.13 synthetic samples. Overall, all these models are lightweight and can be trained on a single GPU. Thus, from a complexity perspective, our method is potentially well-suited for future deployment on personal devices, enabling learning and optimization based on individuals’ ECGs.

CONCLUSION

In this study, we present GAEA-GAN, which leverages rest-exercise ECG pairs from healthy individuals along with resting abnormal ECGs to generate abnormal ECGs enriched with activity-related features. In situations where exercise-induced abnormal ECG data is scarce, our method has the potential to improve the performance of abnormal classification models, mitigating false alarms and missed detections caused by the variability introduced during physical activity. To achieve effective ECG activity enrichment, our separator utilizes a coupled encoder to efficiently disentangle activity-level and health-condition features. The generator, equipped with position-aware instance normalization, enables dynamic injection of activity-related features into the ECG signals. Our extensive experiments on five widely used datasets demonstrate the effectiveness of our method compared to eight state-of-the-art baselines. Additionally, a clinical Turing test involving seven experienced cardiologists further confirms that our

synthesized data exhibit high fidelity, making it challenging to accurately distinguish them from real measured ECGs.

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